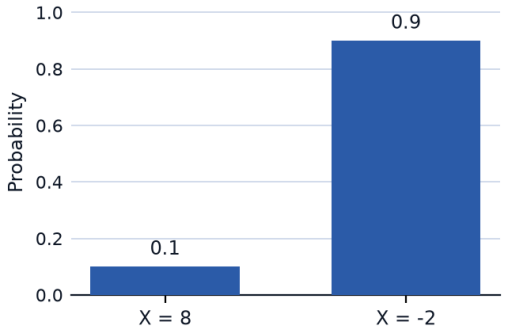
Discrete Random Variables & Binomial Distributions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

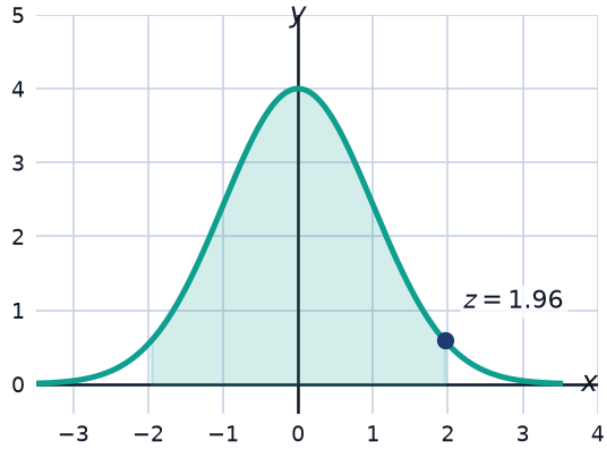
1 [Foundational] • Target: Probability distributions and expected value

A game costs 2 dollars to play. You win 10 dollars with probability 0.1, and win nothing otherwise. Let X be your net profit. Find the probability distribution of X and its expected value.



2 [Foundational] • Target: Probability distributions and expected value

Find the standard deviation of the profit X from the previous problem, to two decimal places.



3 [Foundational] • Target: Binomial setting and conditions

State the four conditions required for a random variable to follow a binomial distribution (sometimes remembered as BINS: Binary, Independent, Number fixed, Same probability).

4 [Foundational] • Target: Binomial setting and conditions

A survey randomly selects 20 people (without replacement) from a small town of 50 people and records how many support a proposal. Explain why this technically violates the independence condition, and why it is often still treated as approximately binomial.

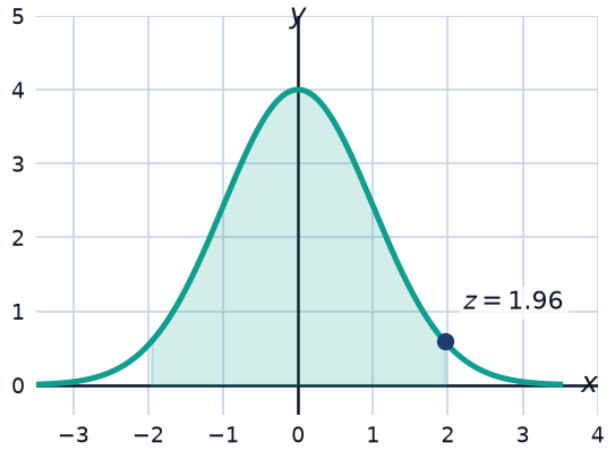
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5 [Foundational] • Target: Binomial probability formula

A basketball player makes 70% of free throws. Find the probability she makes exactly 4 out of 6 attempts, using $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$.

6 [Foundational] • Target: Binomial probability formula

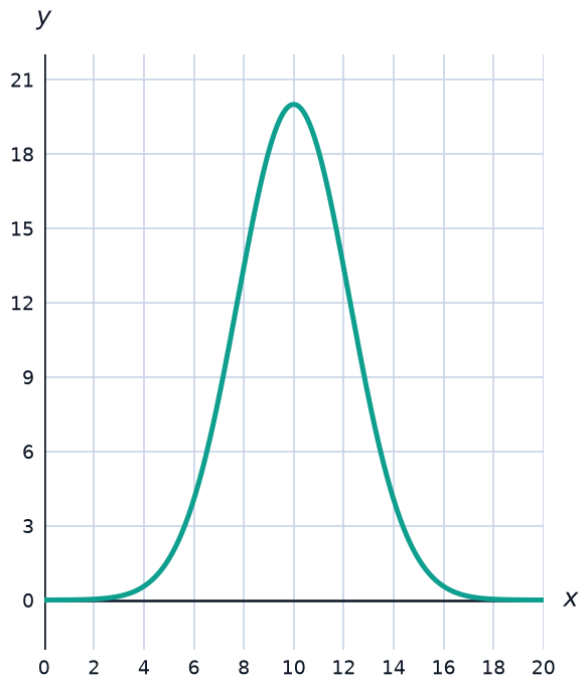
Using the same free-throw shooter, find the mean and standard deviation of the number of makes out of 6 attempts, using $\mu = np$ and $\sigma = \sqrt{np(1-p)}$.



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7 [Foundational] • Target: Visualizing a binomial distribution

The graph shows a smoothed approximation of the binomial distribution for $n = 20$ trials with $p = 0.5$, plotted as a continuous curve for visual comparison to a normal shape.



a Using $\mu = np$ and $\sigma = \sqrt{np(1-p)}$, verify that this binomial distribution has mean 10 and standard deviation approximately 2.24, matching the curve shown.

b Explain why a binomial distribution with $n = 20$ and $p = 0.5$ looks approximately bell-shaped, referencing the conditions for a normal approximation.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Geometric distribution (brief introduction)

A basketball player makes free throws with probability 0.7 each attempt, independently. Find the probability that her first miss occurs on the 3rd attempt (i.e. she makes the first 2, then misses the 3rd).

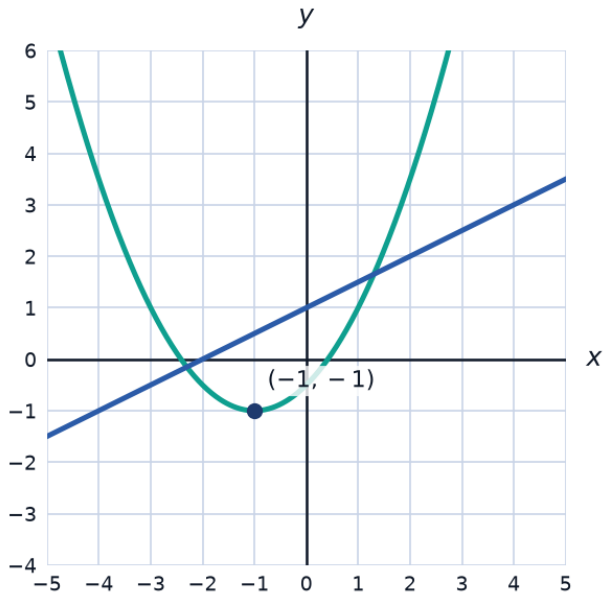
9 [Intermediate] • Target: Applying the binomial distribution in context

A multiple-choice quiz has 10 questions, each with 4 choices, and a student guesses randomly on every question. Find the probability the student gets at least 8 questions correct, given that $P(X = 8) \approx 0.0004$, $P(X = 9) \approx 0.00003$, and $P(X = 10) \approx 0.000001$.

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10 **[Intermediate]** • **Target: Binomial probability of at least one success**

A manufacturing process produces defective items independently with probability 0.05. In a batch of 20 items, find the probability that at least one item is defective, using the complement of "zero defective."



11 **[Intermediate]** • **Target: Binomial probability of at least one success**

Using the mean and standard deviation formulas $\mu = np$ and $\sigma = \sqrt{np(1-p)}$, find the mean and standard deviation of the number of defective items in a batch of 20, with $p = 0.05$.

12 **[Intermediate]** • **Target: Comparing binomial scenarios**

Explain why a binomial model would NOT be appropriate for counting the number of aces dealt when drawing 5 cards without replacement from a standard 52-card deck, and name the distribution that would be more appropriate instead.

13 **[Intermediate]** • **Target: Probability distributions and expected value**

A discrete random variable, X, represents the number of successful trials in a sequence of independent Bernoulli trials. The probability of success on any given trial is $p = 0.6$. The expected value of X is $E[X] = 18$. We are interested in a new random variable, Y, defined as $Y = aX + b$, where a and b are constants. If the variance of Y is $Var(Y) = 7.2$, what is the value of a^2?

14 **[Intermediate]** • **Target: Binomial setting and conditions**

Consider a discrete random variable X that represents the number of heads obtained when a fair coin is tossed 3 times. The probability distribution of X is given by: | x | 0 | 1 | 2 | 3 | |---------|-------|-------|-------|-------| | $P(X = x)$ | $\frac{1}{8}$ | $\frac{3}{8}$ | $\frac{3}{8}$ | $\frac{1}{8}$ | What is the probability that X is less than 2, i.e., $P(X < 2)$?

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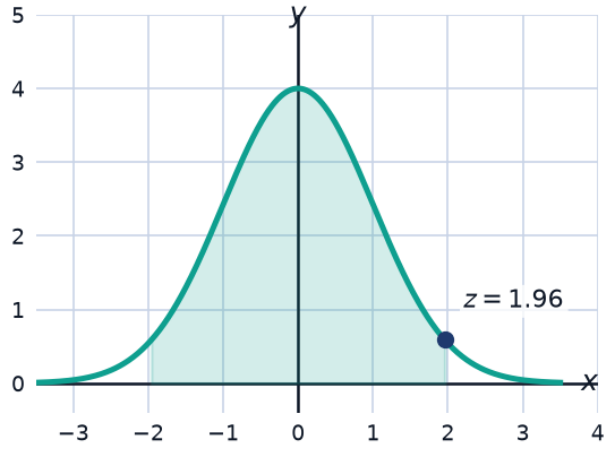
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Binomial probability formula** • **Cross-Lesson: 1.01 Measures of Center & Spread**

A discrete random variable X has probability distribution: $P(X = 0) = 0.1$, $P(X = 1) = 0.3$, $P(X = 2) = 0.4$, $P(X = 3) = 0.2$. Calculate: (a) $E(X)$, (b) $E(X^2)$, and (c) $\text{Var}(X)$ and σ_X.

16 **[Challenging]** • **Target: Visualizing a binomial distribution** • **Cross-Lesson: 1.02 Z-Scores & Linear Transformations**

Independent random variables X and Y have $\mu_X = 50$, $\sigma_X = 6$ and $\mu_Y = 30$, $\sigma_Y = 8$. Define $W = 3X - 2Y + 10$. Find the mean μ_W and standard deviation σ_W.



17 **[Challenging]** • **Target: Geometric distribution (brief introduction)** • **Cross-Lesson: 1.03 Describing Distributions & Linear Regression**

A basketball player makes 75% of free throws. In a game, she shoots 12 free throws. Assuming attempts are independent: (a) Find the probability she makes exactly 9 free throws. (b) Find the mean and standard deviation of made free throws.

18 **[Challenging]** • **Target: Applying the binomial distribution in context** • **Cross-Lesson: 2.01 Sampling Methods & Bias**

A quality control engineer is monitoring the number of defects produced by two different machines, Machine A and Machine B. Let X be the number of defects produced by Machine A in a given hour, and Y be the number of defects produced by Machine B in the same hour. The joint probability distribution of X and Y is given in the table below: | $P(X = x, Y = y)$ | $y = 0$ | $y = 1$ | $y = 2$ | | :----------: | :-------: | :-------: | :-------: | | $x = 0$ | 0.10 | 0.15 | 0.05 | | $x = 1$ | 0.20 | 0.30 | 0.10 | | $x = 2$ | 0.05 | 0.05 | 0.00 | What is the expected value of the total number of defects produced by both machines in a given hour, $E(X + Y)$?

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19 [Challenging] • Target: Binomial probability of at least one success • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

A random sample of 200 students is classified by Grade Level (9th vs 12th) and Smartphone Operating System (iOS vs Android). Observed counts: 9th Grade: 70 iOS, 30 Android (100 total); 12th Grade: 50 iOS, 50 Android (100 total). (a) Calculate the expected count of 9th graders with iOS. (b) Compute the Chi-square test statistic χ^2. (c) With $df = 1$ and $\alpha = 0.05$ (critical value 3.84), conclude whether Grade Level and Operating System are independent.

20 [Challenging] • Target: Comparing binomial scenarios • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

In an SRS of 200 registered voters, 116 favor a new environmental regulation. (a) Check all three conditions for constructing a 95% confidence interval for the population proportion. (b) Construct the 95% confidence interval. (c) Interpret the interval in context.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Probability distributions and expected value

A binomial random variable has $n = 14$ trials with success probability $p = 0.7$. Find the mean $\mu = np$.

a. 4.2

b. 10.8

c. 8.8

d. 9.8

22 [Foundational] • Target: Binomial setting and conditions

A binomial random variable has $n = 30$ trials with success probability $p = 0.3$. Find the mean $\mu = np$.

a. 10.0

b. 8.0

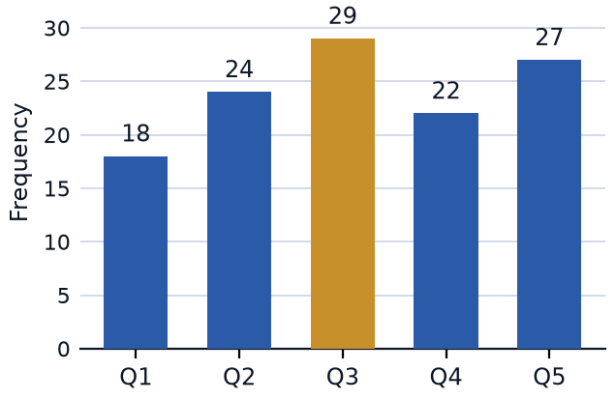
c. 9.0

d. 21.0

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23 [Foundational] • Target: Binomial probability formula

A binomial random variable has $n = 36$ trials with success probability $p = 0.6$. Find the mean $\mu = np$.



a. 21.6

b. 22.6

c. 20.6

d. 14.4

24 [Foundational] • Target: Visualizing a binomial distribution

A binomial random variable has $n = 21$ trials with success probability $p = 0.4$. Find the mean $\mu = np$.

a. 12.6

b. 8.4

c. 7.4

d. 9.4

25 [Foundational] • Target: Geometric distribution (brief introduction)

A binomial random variable has $n = 30$ trials with success probability $p = 0.4$. Find the mean $\mu = np$.

a. 13.0

b. 12.0

c. 11.0

d. 18.0

26 [Foundational] • Target: Applying the binomial distribution in context

A binomial random variable has $n = 13$ trials with success probability $p = 0.3$. Find the mean $\mu = np$.

a. 2.9

b. 4.9

c. 3.9

d. 9.1

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27 [Foundational] • Target: Binomial probability of at least one success

A binomial random variable has $n = 25$ trials with success probability $p = 0.6$. Find the mean $\mu = np$.

a. 16.0

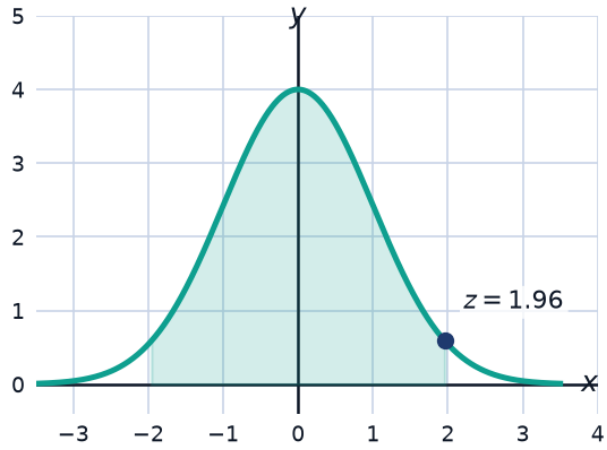
b. 14.0

c. 10.0

d. 15.0

28 [Foundational] • Target: Comparing binomial scenarios

A binomial random variable has $n = 15$ trials with success probability $p = 0.9$. Find the standard deviation $\sigma = \sqrt{np(1-p)}$, to two decimal places.



a. 1.35

b. 1.16

c. 0.86

d. 1.46

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Expected Value of a Discrete Random Variable

A binomial random variable has $n = 29$ trials with success probability $p = 0.6$. Find the standard deviation $\sigma = \sqrt{np(1-p)}$, to two decimal places.

a. 2.34

b. 6.96

c. 2.64

d. 2.94

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30 [Intermediate] • Target: Variance & Standard Deviation of Combinations

A binomial random variable has $n = 34$ trials with success probability $p = 0.8$. Find the standard deviation $\sigma = \sqrt{np(1-p)}$, to two decimal places.

a. 2.63

b. 2.33

c. 5.44

d. 2.03

31 [Intermediate] • Target: The Binomial Setting & Probability Formula

A binomial random variable has $n = 33$ trials with success probability $p = 0.1$. Find the standard deviation $\sigma = \sqrt{np(1-p)}$, to two decimal places.

a. 2.02

b. 1.42

c. 2.97

d. 1.72

32 [Intermediate] • Target: Mean & Standard Deviation of Binomial Distributions

A binomial random variable has $n = 11$ trials with success probability $p = 0.6$. Find the standard deviation $\sigma = \sqrt{np(1-p)}$, to two decimal places.

a. 1.62

b. 1.32

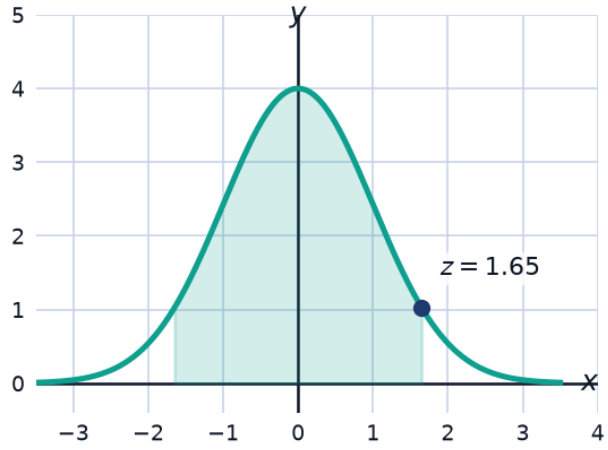
c. 1.92

d. 2.64

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33 [Intermediate] • Target: Probability distributions and expected value

A binomial random variable has $n = 39$ trials with success probability $p = 0.8$. Find the standard deviation $\sigma = \sqrt{np(1-p)}$, to two decimal places.



a. 2.50

b. 2.80

c. 2.20

d. 6.24

34 [Intermediate] • Target: Binomial setting and conditions

A binomial random variable has $n = 24$ trials with success probability $p = 0.9$. Find the standard deviation $\sigma = \sqrt{np(1-p)}$, to two decimal places.

a. 2.16

b. 1.77

c. 1.47

d. 1.17

35 [Intermediate] • Target: Binomial probability formula

A manufacturing process produces defective items independently with probability 0.1. In a batch of 14 items, find the probability that at least one item is defective, using the complement of "zero defective," to three decimal places.

a. 0.721

b. 0.771

c. 0.821

d. 0.229

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36 [Intermediate] • Target: Visualizing a binomial distribution

A manufacturing process produces defective items independently with probability 0.1. In a batch of 12 items, find the probability that at least one item is defective, using the complement of "zero defective," to three decimal places.

a. 0.282

b. 0.668

c. 0.718

d. 0.768

37 [Intermediate] • Target: Geometric distribution (brief introduction)

A process succeeds independently with probability $p = 0.8$ on each of $n = 6$ trials. Find the probability of exactly 2 successes, using $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$, to three decimal places.

a. 0.031

b. 0.015

c. 0.001

d. 0.065

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Applying the binomial distribution in context • Cross-Lesson: 2.01 Sampling Methods & Bias

A process succeeds independently with probability $p = 0.2$ on each of $n = 8$ trials. Find the probability of exactly 1 successes, using $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$, to three decimal places.

a. 0.286

b. 0.386

c. 0.671

d. 0.336

39 [Challenging] • Target: Binomial probability of at least one success • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

A game costs 10 dollars to play. You win 31 dollars with probability 0.3, and win nothing otherwise. Letting X be your net profit, find $E(X)$.

a. $-\frac{17}{10}$

b. $\frac{3}{10}$

c. $-\frac{7}{10}$

d. $\frac{117}{10}$

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40 [Challenging] • Target: Comparing binomial scenarios • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

A game costs 9 dollars to play. You win 20 dollars with probability 0.5, and win nothing otherwise. Letting X be your net profit, find $E(X)$.

a. $\frac{3}{2}$

b. 1

c. 2

d. 0

41 [Challenging] • Target: Expected Value of a Discrete Random Variable • Cross-Lesson: 3.03 The Normal Distribution

A discrete random variable X has distribution: $P(X=0)=0.2, P(X=1)=0.5, P(X=2)=0.3$. Find $E(X)$.

a. 1.1

b. 1.0

c. 1.5

d. 0.8

42 [Challenging] • Target: Expected Value of a Discrete Random Variable • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

If $E(X)=12$ and $Y=3X-5$, what is $E(Y)$?

a. 31

b. 36

c. 41

d. 12

43 [Challenging] • Target: Variance & Standard Deviation of Combinations • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

Why must variances be added rather than standard deviations when combining independent variables?

a. Standard deviations are not in squared units, while the Pythagorean relationship of orthogonal independent variables governs variances

b. Standard deviations can be negative

c. Adding standard deviations always gives an imaginary number

d. Because variance is always smaller than standard deviation

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44 [Challenging] • Target: The Binomial Setting & Probability Formula • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A fair 6-sided die is rolled 5 times. What is the probability of rolling exactly two 4s?

a. $\binom{5}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^3$

b. $\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^3$

c. $\binom{5}{2}\left(\frac{1}{6}\right)^3\left(\frac{5}{6}\right)^2$

d. $\frac{2}{5}$

45 [Challenging] • Target: Mean & Standard Deviation of Binomial Distributions • Cross-Lesson: 1.01 Measures of Center & Spread

A multiple-choice test has 20 questions, each with 4 options. If a student guesses completely at random, what are the mean and standard deviation of correct answers?

a. $\mu = 5.0, \sigma = 1.94$

b. $\mu = 5.0, \sigma = 3.75$

c. $\mu = 4.0, \sigma = 2.00$

d. $\mu = 5.0, \sigma = 15.0$

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